

```
# Logistic Regression Algorithm
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```
# Import libraries
```

```
from sklearn.datasets import load_iris
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.metrics import accuracy_score, confusion_matrix
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
# Load Iris dataset
```

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

```
# Split dataset into training and testing
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42
```

```
)
```

```
# Create Logistic Regression model
```

```
model = LogisticRegression(max_iter=200)
```

```
# Train the model
```

```
model.fit(X_train, y_train)
```

```
# Predict test data
```

```
y_pred = model.predict(X_test)
```

```
# Calculate accuracy
```

```
accuracy = accuracy_score(y_test, y_pred)

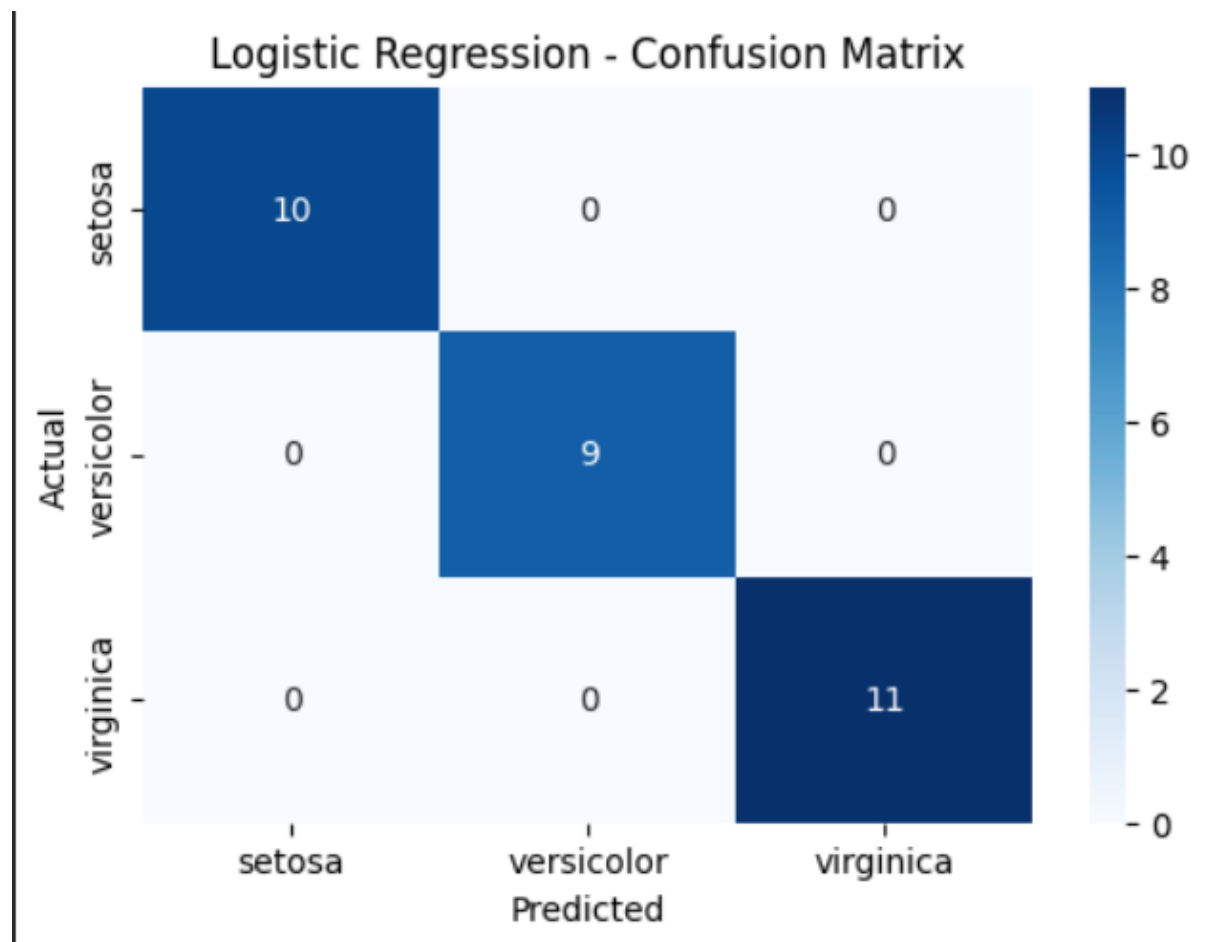
print("Logistic Regression")
print("-----")
print("Accuracy:", accuracy)
print("Accuracy (%)ate", accuracy * 100)

# Create confusion matrix
cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

# Display confusion matrix
plt.figure(figsize=(6, 4))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=iris.target_names,
    yticklabels=iris.target_names
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Logistic Regression - Confusion Matrix")
plt.show()
```



Logistic Regression

Accuracy: 1.0

Accuracy (%): 100.0

Confusion Matrix:

```
[10  0  0]
[ 0  9  0]
[ 0  0 11]
```